Coupling the Point Model to the Growth Field

Key points of the meeting of 20 October 2026

Keisuke Nishioka

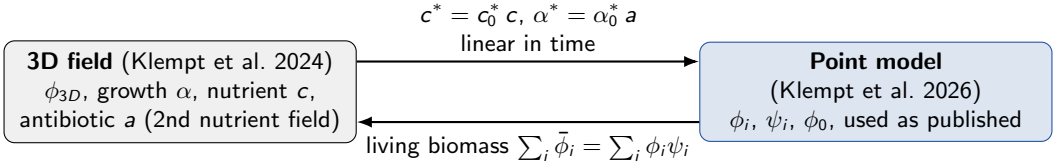
Institute of Continuum Mechanics (IKM), Leibniz Universität Hannover

20 October 2026

What changes

- Point model (Klempt et al. 2026) used as published
- No scaling to the field ($\phi_0 = 1 - \phi_{3D}$ dropped)
- Field $\rightarrow$ point model: c^*, α^* from 3D fields
- Point model $\rightarrow$ field: only living biomass consumes
- Eq. 34 with front term, Eq. 36 unchanged

The loop at one Gauss point



$\sum_i \bar{\phi}_i$ scales the consumption of c and a $\rightarrow$ c enters the front term of Eq. 34 $\rightarrow$ spreading of ϕ_{3D} $\rightarrow$ growth α (Eq. 36) $\rightarrow$ stress

Equations

$$c^*(t) = c_0^*\left[\tfrac{t_{n+1}-t}{\Delta t}c_n + \tfrac{t-t_n}{\Delta t}c_{n+1}\right], \quad \alpha^*(t) = \alpha_0^*\left[\tfrac{t_{n+1}-t}{\Delta t}a_n + \tfrac{t-t_n}{\Delta t}a_{n+1}\right]$$

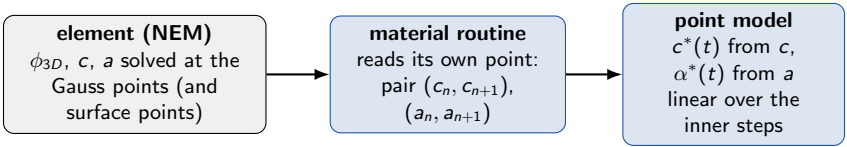
$$\dot{c} = d\,\nabla^2 c - g\,\phi_{3D}\,\textstyle\sum_i \bar{\phi}_i\, c, \qquad \bar{\phi}_i = \phi_i\psi_i$$

$$\dot{a} = d_a\nabla^2 a - g_a\,\phi_{3D}\,\textstyle\sum_i \bar{\phi}_i\, a$$

$$\dot{\phi}_{3D} = \beta\nabla^2\phi_{3D} + k_\alpha\alpha - \|\nabla\phi_{3D}\|\,\tfrac{r\,c}{k+c}\,\mathbf{n}_{\nabla\phi_{3D}}\cdot\mathbf{n}_{\nabla c}, \qquad \dot{\alpha} = k_\alpha\phi_{3D}$$

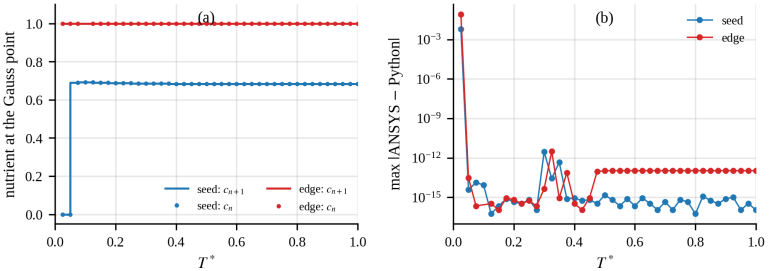
$g\phi$ in Eq. 35 of the paper is a misprint; $g\phi_{3D}c$ is used (Monod form $f(c) = c/(K_c + c) \to c/K_c$ for $K_c \gg c$). ϕ_{3D}: field, ϕ_i: point model. Antibiotic: a (field), α^* (point model); growth: α (Eq. 36), not passed to the point model.

Nutrient and antibiotic at the Gauss points



- No interpolation from nodes: the fields live at the integration points
- The antibiotic in the second nutrient field takes the same path
- One increment behind: the fields are solved after the material routine
- A short first load step solves the fields before the point model starts

First ANSYS run: what is checked



- (a) c_{n+1} of one increment is c_n of the next: the pair is passed on without a jump
- (b) point model in ANSYS = Python with the same inputs: 2×10^{-12} or better in every step (four runs, with and without antibiotic)
- The first step needs up to 38 Newton iterations from the start state; with a tolerance-controlled Newton (up to 50) both codes agree there too

Time: one clock for both models

- $T^* = 1$ = duration of the point model's data
- s follows from it, not fitted (synthetic: $s = 0.1$)
- Check (synthetic): error 7×10^{-7}; with scaling 0.019

Examples and checks

Examples (synthetic, two species)

- A centre, B edges
- A and B mixed
- with / without antibiotic

Checks

- equals point model alone
- ANSYS = Python
- $0 \leq \phi \leq 1$
- ϕ_{min}, step, mesh

Results to follow

Assumptions and open points

- Assumptions: linear interpolation, $\sum_i \bar{\phi}_i$ one step late, start values of new points, ϕ_{min}
- No transport of species between points
- Front term needs upwinding
- Antibiotic: synthetic data only ($b = 0$ in the calibration)

Appendix: notation

ϕ_{3D}, α	biofilm phase field (ϕ of Klempt et al. 2024) and growth ($\mathbf{F}_g = \alpha\mathbf{I}$, $\alpha(0) = 1$), Klempt et al. 2024
c, a	nutrient and antibiotic fields (the element's first and second nutrient)
d, g, d_a, g_a	diffusion and consumption of nutrient and antibiotic
β, k_α, r, k	diffusion of ϕ, growth rate, front growth and Monod constant (Eq. 34, 36)
ϕ_i, ψ_i, ϕ_0	volume fraction and living fraction of species i, auxiliary void (Klempt et al. 2026)
c^*, α^*, b_i	point-model nutrient $c_0^* c$, antibiotic $\alpha_0^* a$ (not the growth α); sensitivity
$\bar{\phi}_i$	living volume fraction of species i, $\phi_i \psi_i$ (Klempt et al. 2026)
T^*, s	normalised time; point-model time per unit T^*